Peiyu Zhang

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Education

University of Virginia, PhD Candidate of Systems and Information Engineering — Sep 2022 – Present

- **Research Interests: Human-Computer Interaction, Artificial Intelligence, Human Factors Engineering**
- GPA: 4.0/4.0, MEng obtained: May 2025 | PhD expected: May 2027

University of Virginia, BA in Computer Science with High Distinction — Sep 2020 – May 2022

- GPA: 4.0/4.0, Minor: Psychology

Technical Skills

Languages & Frameworks: Python, **Java**, **C++**, C#, SQL, Kotlin, JavaScript, **PyTorch**, React, Node.js

Tools & Platforms: LangChain, Apache Spark, AWS, AI Model Platforms & APIs, MongoDB, Unity, Android Studio

Experience

Pony.ai, PhD Software Engineering Intern (ADV Behavior) — Aug. 2026 – Oct. 2026

- Design, evaluate, and improve **decision-making logic for L4 autonomous driving systems**, supporting vehicle behavior planning across complex driving scenarios.
- Develop and validate **performance metrics to assess autonomous driving behavior** both onboard and offline, facilitating systematic evaluation and continuous improvement.
- Optimize decision-system **code flows and software architecture in C++**, improving logic organization, onboard latency, and maintainability.
- Configure and improve simulation services in **Kubernetes-based** development and deployment environments.

University of Virginia Riggs' Lab, Graduate Research Assistant — Jan. 2021 – Present

- Built multiple **VR/AR applications** using **Unity (C#)** and **Android Studio** for **Meta Quest**, **Microsoft HoloLens**, and **Magic Leap**, supporting **multimodal input/output testing** and interactive task-switching environments.
- Led a series of mixed reality experiments analyzing cognitive workload and multi-interface attention through **multimodal data analysis** of eye tracking, movement tracking, and event logs using **Python** and **R** for **statistical modeling and visualization**.
- Designed human-in-the-loop evaluations of AI-assisted clinical decision-making and **human–AI collaboration** workflows, and iterated on **AI system design** for diagnostic/visualization tools to enhance performance and promote calibrated trust.
- Explored **NLP/LLM** approaches for attitude/stance understanding in online discussion communities, and prototyped **LLM-agent–based** therapeutic assistance tools and VR interactive systems.
- Developed a VR application to record and **process 3D movement data**, integrating real-time sensor input and spatial tracking for **quantitative kinematic analysis**.

University of Virginia Ultimate User Interface Lab, Graduate Research Assistant — Sep 2021 – Sep 2023

- Developed a **computer vision–based wearable sensing system** to detect micro-movements and physical changes (e.g., force, volume) using **Python**, **PyTorch**, **OpenCV**, and convolutional neural networks **(CNNs)** informed by the Moiré pattern optical effect.
- Built **processing pipelines** for motion tracking and trained image recognition models to detect **real-time physical interactions**, enabling tangible interfaces via spatial gesture input, and integrated the system into self-developed **Android** and **VR applications**.
- Contributed to experimental design and VR prototyping to investigate VR-based perception manipulation in reaching tasks using a **LSTM autoencoder** for **time series data analysis**.

University of Virginia Physics Department, Graduate Research Assistant — May 2022 - Sep 2022

- Created an interactive VR system in Unity for immersive science education, engineering 3D models and integrating physics-based animation pipelines to visualize real-time parton dynamics.

Software Projects

AI-Powered Personal Finance Dashboard

- Developed a full-stack personal finance dashboard using **React**, **Node.js**, and **MongoDB**, allowing users to log expenses and investments, manage budgets, and track real-time stock holdings.
- Integrated live stock market data from the **Finnhub API**, enabling users to monitor stock performance, view gains and losses, and explore historical price trends.
- Built an AI-powered insights module using the **OpenAI API** to generate natural language summaries of spending habits, personalized saving tips, and stock-related news and suggestions.
- Built interactive visualizations with **Recharts** for spending, stock trends, and budget-vs-actual comparisons.

AI Agentic Travel Planner

- Developed a **LangChain-based multi-agent planner** (Planner, Verifier, Local Guide, Budgeter) to generate itineraries, validate constraints, and produce cost breakdowns.
- Built a **RAG pipeline** over personal travel notes and itineraries to ground recommendations, and created a "travel taste" **embedding profile** from past trips and saved places to rank suggestions that match user style.
- Implemented constraint-check tools and safety guardrails to reduce errors and improve trustworthiness.

Android AR Treasure Hunt Mobile Game

- Developed an AR-based Android game using **ARCore** and **Android Studio**, allowing users to explore physical environments and discover virtual collectibles overlaid in real space.
- Implemented plane detection and hit testing with ARCore to anchor virtual game objects (e.g., coins, gold bars) onto real-world surfaces in real time.
- Implemented real-time **UDP communication** for local multiplayer gameplay, allowing devices on the same network to broadcast player actions and synchronize game state.
- Designed a modular game logic system using **Kotlin**, featuring progressive difficulty, interactive animations, scorekeeping, collectible tracking, and real-time visual feedback to enhance player engagement.
- Integrated persistent game state using Room **(SQLite)** to store player progress, collected items, and scores across sessions.

Publications

A Signal Detection Analysis of Visuo-Tactile Dual-Tasking Performance Under Varying Temporal Task Loads — Oct 2026

Mohamad El Iskandarani, ***Peiyu Zhang***, Matthew L. Bolton, and Sara L. Riggs

HFES Annual Meeting (ASPIRE) 2026, DOI: 10.1177/10711813261498787

Evaluating Gaze-Responsive Visual Effects for Multi-Window Visual Search in Virtual Interfaces — Oct 2026

Peiyu Zhang, Mohamad El Iskandarani, and Sara L. Riggs

HFES Annual Meeting (ASPIRE) 2026, DOI: 10.1177/10711813261493584

Web Pages as 3D Proxies in VR: A Design Framing and Preliminary Exploration — Apr 2026

Tianjie Shu* and ***Peiyu Zhang****

ACM CHI EA 2026, DOI: 10.1145/3772363.3798605

Multitasking Across Physical and Virtual Displays: The Effect of Spatial Discontinuity and Task Load — Mar 2026

🏆 **Honorable Mention Award**

Peiyu Zhang, Mohamad El Iskandarani, and Sara L. Riggs

IEEE VR 2026 (Special Issue of IEEE TVCG), DOI: 10.1109/TVCG.2026.3679118

The Effect of Layout on Visual Search in Augmented Reality Multi-Window Displays — Dec 2025

Peiyu Zhang, Mohamad El Iskandarani, and Sara L. Riggs

IEEE Transactions on Visualization and Computer Graphics, DOI: 10.1109/TVCG.2025.3642044

The Effect of AI Assistance Timing on Performance and User Perceptions in Pathological Slide Diagnosis Oct 2025

Peiyu Zhang, Zichen Ye, Ronggan Wei, Haiyan Niu, Hongxia Li, Mingjuan Wang, Peng Xue, and Sara L. Riggs

Applied Ergonomics, DOI: 10.1016/j.apergo.2025.104663

Deep learning-assisted system improves practical effect in cervical cytopathology diagnosis: A comparative study of reading modes Sep 2025

Zichen Ye, ***Peiyu Zhang***, Ronggan Wei, Haiyan Niu, Hongxia Li, Mingjuan Wang, Sara L. Riggs, and Peng Xue

Modern Pathology, DOI: 10.1016/j.modpat.2025.100882

Multitasking Between Realities: An Examination of Multitasking Using Augmented Reality Under Different Task Demand Levels Jul 2025

Peiyu Zhang and Sara L. Riggs

HFES Annual Meeting (ASPIRE) 2025, DOI: 10.1177/10711813251357901

MoiréTag: A Low-Cost Tag for High-Precision Tangible Interactions without Active Components Oct 2024

🏆 **Honorable Mention Award (Top 3 Papers)**

Peiyu Zhang, Wen Ying, Sara L. Riggs, and Seongkook Heo

ACM ISS 2024, DOI: 10.1145/3698113

Chinese colposcopists' attitudes toward the colposcopic artificial intelligence auxiliary diagnostic system (CAIADS): A nation-wide, multi-center survey Sep 2024

Huike Wang*, Zichen Ye*, ***Peiyu Zhang****, Xiaoli Cui, Mingyang Chen, Aiyuan Wu, Sara L. Riggs, Peng Xue, and Youlin Qiao

Sage Digital Health, DOI: 10.1177/20552076241279952

Fringer: A Finger-Worn Passive Device Enabling Computer Vision Based Force Sensing Using Moiré Fringes Oct 2022

Peiyu Zhang, Wen Ying, and Seongkook Heo

UIST 2022, DOI: 10.1145/3526114.3558706